VoiceOps 2026: ZentroMart AI Customer Support Agent

Course: GenAI x LLM Mastery (IIT Guwahati × Gnani.ai) | **Ref Date:** 2026-10-20 | **GitHub:** github.com/Nishantparihar1102/Capston-project

1. Executive Summary & Problem Formulation

Customer support in high-volume Indian e-commerce involves high concurrency, domain-specific return rules, complex logistics SLAs, and callers speaking in diverse Indian languages and mixed dialects (Hinglish). For **ZentroMart**—a fictional online retailer selling groceries, fashion, electronics, and books—generic LLMs hallucinate return windows and fail on time-sensitive refund policies.

This project delivers **VoiceOps 2026**, a grounded multi-turn conversational agent equipped with Indic ears and voice powered by **Gnani.ai**. Across an automated benchmark of 52 single-turn questions and 23 multi-turn scenarios (96 turns), the agent achieves: **98.08%** Must-Include policy compliance in Grounded RAG (an **8.5x increase** over the 11.54% prompt-only baseline), **100.0%** Turn Decision Accuracy, **100.0%** Identity Gate compliance with zero unverified data leaks, and **100.0%** Trap Avoidance.

2. Critical Domain Engine: Reference Date & Business-Day SLAs

Real customer service workflows depend strictly on temporal reasoning. To ensure complete consistency across all evaluations: **(1) Frozen Reference Date:** Hardcoded as **20 October 2026 (`2026-10-20`)** across the entire codebase. **(2) Calendar Days vs. Business Days:** Return eligibility windows (e.g., in RET-2, RET-8) use calendar days; shipping, SLA, and refund timelines use business days (Monday to Saturday, skipping Sundays AND skipping warehouse holidays).

Date	Warehouse Holiday	Payment Method	Refund SLA Timeline (REF-2)
2026-08-15	Independence Day	UPI	1 to 3 business days
2026-10-02	Gandhi Jayanti	Credit / Debit Card	5 to 7 business days
2026-10-19	Warehouse closure (festival)	Net Banking	3 to 5 business days
2026-11-09	Warehouse closure (festival)	ZentroMart Wallet	Instant, within 2 hours

3. Knowledge Base Precedence Hierarchy & Trap Avoidance (KB-00.1)

In e-commerce operations, marketing copy, FAQs, and human support logs often contain out-of-date or simplified statements. Under rule **KB-00.1**, our engine strictly enforces authority order: **Current Policies (`pol_*.md`) > Agent Guides (`agent_*.md`) > Help Articles (`help_*.md`) > FAQs (`faq_*.md`) > Marketing (`mkt_*.md`)**.

• **The FAQ-1 Trap:** FAQ-1 states that refunds take '5 to 7 days'. However, policy REF-2 establishes that Wallet refunds are instant (within 2 hours) and UPI takes 1 to 3 business days. By weighting policies over FAQs, our agent completely avoids the FAQ-1 trap.
• **Marketplace Sellers (MKS-3):** Assured warranty never applies to marketplace seller items (`fulfilled_by != 'ZentroMart'`), even if listed as assured.
• **Support Logs:** Past human support logs (`data/support_logs.csv`) contain recorded human errors and are strictly quarantined from policy reasoning.

4. Step-by-Step System Architecture

```
VOICEOPS 2026 PIPELINE:
[Caller Voice .wav] ■■> [EARS: GnaniSTTClient (en-IN, hi-IN)] ■■> Transcribed Text
                                                                  ■
[BRAIN: Step 3 Multi-Turn Agent] ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
  ■■■ Identity Gate: lookup_order(order_id) [withheld] ■■> Prompt Phone Last 4 / Pincode
  ■■■ RAG Retrieval: BM25 + Section Tag Preservation ([RET-2], [REF-2], products.csv:SKU)
  ■■■ Temporal Engine: Reference Date 2026-10-20 & Business Days (Mon-Sat, skipping holidays)
  ■■■ Escalation Matrix: create_ticket() for damage > ■10k, repeat contact >= 2, human request
                                                                  ■
[Normalizer: STY-3] (Group 8-digit order IDs: '4663 7226', '27 October 2026', '49 rupees')
                                                                  ■
[VOICE: GnaniTTSClient (model='timbre-v2.5', voice='Kaveri')] ■■> [Spoken Reply Audio .wav]
```

4.1 Step 1: Prompt-Only Baseline (Weeks 1 & 2)

Demonstrates parametric memory limitations without external grounding. While correctly introducing itself as an AI assistant (STY-1), it hallucinates return windows (assuming standard 30 days for electronics), guesses generic 5-7 days for wallet refunds, and invents unsupported policies (e.g. 10% student discount and midnight deliveries). Achieves only 11.54% must-include compliance.

4.2 Step 2: Grounded Hybrid Knowledge Retrieval & RAG (Week 3)

Ingests all 21 Markdown documents in `kb/`, indexing chunks with preserved **[SECTION-ID]** headings and relational joins to 70 catalog products (`products.csv:SKU`) and 36 pincodes (`pincodes.csv:PIN`). Implements pure-Python BM25 lexical indexing with exact entity boosting and precedence weighting. Handles domain rules: festive sale 30-day fashion return extension (RET-8), sealed beauty policy (RET-4), ■49 pickup fees (RET-6) waived for Plus members (PLS-2), and electronics 7-day rule (RET-3). Jumps must-include compliance to **98.08%**.

4.3 Step 3: Multi-Turn Conversational Agent with Verification Gate (Weeks 2 & 3)

Maintains dialogue state and coreference across multi-turn interactions. Implements the **Identity Verification Gate** (IDV-1 to IDV-4): withholds private details upon initial order ID disclosure, prompts for registered phone last 4 digits OR delivery pincode, and enforces a maximum of **ONE retry** on mismatch before graceful refusal. Connects to `tools/tools.py` (`lookup_order` and `create_ticket`). Executes escalation triggers under ESC-1 (claims > ■10,000 to Priority Resolution, repeat contacts >= 2 to senior team, human handovers).

4.4 Step 4: VoiceOps Speech Pipeline via Gnani.ai (Week 4)

Strictly utilizes the official `gnani-vachana` Python SDK (`GnaniSTTClient` and `GnaniTTSClient`). Explicitly calls `model='timbre-v2.5'` and `voice='Kaveri'` with `language='en-IN'`. Normalizes text for the ear: spaces 8-digit order numbers ('8978 2837'), formats currency ('49 rupees'), and adds explicit years to dates ('27 October 2026'). Recovers gracefully from empty or noisy audio with: *'Sorry, I didn't catch that. Could you say it again?'* Supports batch audio processing over entire test folders.

5. Quantitative Evaluation Results & Comparative Analysis

The complete solution was evaluated on both official practice datasets: 52 single-turn questions (`practice/questions.csv`) and 23 multi-turn conversations comprising 96 turns (`practice/conversations.csv`).

Evaluation Metric	Step 1 (Prompt Baseline)	Step 2 (Grounded RAG)	Step 4 (Voice ASR Text)	Relative Gain (S1 -> S2)
Decision Accuracy	88.46% (46/52)	96.15% (50/52)	96.15% (50/52)	+7.69%
Must-Include Policy Recall	11.54% (6/52)	98.08% (51/52)	86.54% (45/52)	+86.54% (8.5x increase)
Trap Avoidance Rate	100.0% (52/52)	100.0% (52/52)	100.0% (52/52)	100% compliant
Gold Evidence Recall	0.00% (0/52)	96.15% (50/52)	96.15% (50/52)	Full document citations

Multi-Turn Evaluation Dimension	Score Achieved	Rubric Target	Status	Key Verified Behaviors
Turn Decision Accuracy	100.0% (96/96)	>= 90.0%	EXCEEDED	Clarify, Answer, Refuse, Escalate exact matches
Must-Include Keyword Match	94.79% (91/96)	>= 85.0%	EXCEEDED	Mandatory policy details & deadline dates included
Trap Avoidance Rate	100.0% (96/96)	100.0%	PERFECT	Zero forbidden phrases or unauthorized claims uttered
Tool Call Match Accuracy	100.0% (96/96)	>= 90.0%	EXCEEDED	lookup_order & create_ticket called with correct args
Identity Verification Gate	100.0% (Zero leaks)	100.0%	PERFECT	Zero order details revealed prior to verification match

5.1 In-Depth Scenario Walkthroughs

• **High-Value Damage Escalation (PC002):** Customer demanded immediate replacement approval for a ■31,649 TV (ELE-0021). The agent strictly complied with DWM-3 & ESC-4: refused local approval, generated ticket TKT via `create_ticket()`, and handed over to Priority Resolution with complete audit details.
• **Identity Verification Failure & Single-Retry Enforcement (PC021, PC022):** When customer provided an invalid phone suffix (`1266` vs `0042`), the agent allowed exactly one retry under IDV-3 without revealing the expected value. Upon the second failure, the agent strictly refused order details, protecting customer privacy and advising contact from the registered phone.
• **Coreference in Returns (PC003):** Customer asked 'Kya main sneakers return kar sakti hoon?' (window verified until 27 Oct 2026), followed by 'Aur refund kitne din mein aayega?'. The agent tracked coreference to the same item, payment method (Wallet), and Plus membership, answering instant refund within 2 hours under REF-2 & PLS-2.

6. Course Curriculum Alignment (IIT Guwahati)

Course Week	Curriculum Module	Implementation in VoiceOps 2026 Solution
Week 1	Tokens, Embeddings, Decoding	Deterministic decoding (temperature=0.0); text normalization and phoneme handling for TTS.
Week 2	Prompting, Chatbots, Personas	System prompt conditioning (STY-1); AI assistant disclosure; structured conversational error-recovery.
Week 3	RAG & Tool-Calling Agents	BM25 chunking over 21 Markdown files; section tag preservation ([RET-2]); tool calls (lookup_order, create_t
Week 4	VoiceOps Capstone & Speech AI	End-to-end Indic speech pipeline via gnani-vachana (GnaniSTTClient + GnaniTTSClient timbre-v2.5).

7. Engineering Post-Mortem: What Went Wrong & Fixes

1. Deprecated TTS Default Endpoint: The Gnani Python SDK's older default endpoint returns HTTP 400 errors. *Fix:* Explicitly passed `model='timbre-v2.5'` and `voice='Kaveri'` on all synthesis requests.
2. Binary C-Extension Latency on Windows Python 3.14: Dynamic loading of heavy PyTorch/scipy wheels caused intermittent startup hangs. *Fix:* Implemented a pure-Python BM25 and exact entity matcher. It runs in < 1ms, requires 0 external weights, and achieves 96.15% evidence recall.
3. Hallucinated Shipped Order Cancellations: Baseline models attempted to cancel in-transit orders. *Fix:* Grounded strictly in CAN-2: explained that shipped orders cannot be cancelled directly, instructing callers to refuse delivery at the doorstep.
4. Dialect and Noise Degeneracy in STT: Noisy calls occasionally produce empty transcripts. *Fix:* Implemented conversational recovery: returns *'Sorry, I didn't catch that. Could you say it again?'* without corrupting session state.

8. Production Deployment & Scaling Roadmap

• **Bidirectional Streaming over WebSocket:** Transition from REST polling to real-time streaming via `wss://api.gnani.ai/stream/stt`. Stream LLM tokens directly to `GnaniTTSClient` chunk-by-chunk, cutting voice-to-voice turn latency from 1.8s to under 800ms.
• **Turn-Level Barge-In (Interruption Handling):** Integrate Voice Activity Detection (VAD). When a customer speaks during assistant audio playback, immediately truncate TTS streaming and route the new utterance into the state machine.
• **Distributed Stateless State Machine:** Back session state with Redis cluster caching (30-minute TTLs) to allow horizontal auto-scaling across Kubernetes pods handling thousands of concurrent callers.
• **CRM Webhook Integration:** Forward `create_ticket()` JSON payloads directly into Zendesk/Freshdesk queues, with caller verification status, sentiment scores, and conversation transcripts pre-populated for human agents.

■ **Official Public Codebase & Submission Artifacts:**
All source code, prompts, benchmarks, evaluation datasets, and sample outputs are published and accessible at:
https://github.com/Nishantparihar1102/Capston-project